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(54) **SECURITY / AUTOMATION SYSTEM
CONTROL PANEL WITH SHORT RANGE
COMMUNICATION DISARMING**

(71) Applicant: **Johnson Controls Tyco IP Holdings
LLP**, Milwaukee, WI (US)

(72) Inventors: **Walter Wallach**, San Jose, CA (US);
Anand Sastry, Los Altos Hills, CA
(US); **David Pulling**, San Jose, CA
(US)

(73) Assignee: **Tyco Fire & Security GmbH**,
Neuhausen am Rheinfall (CH)

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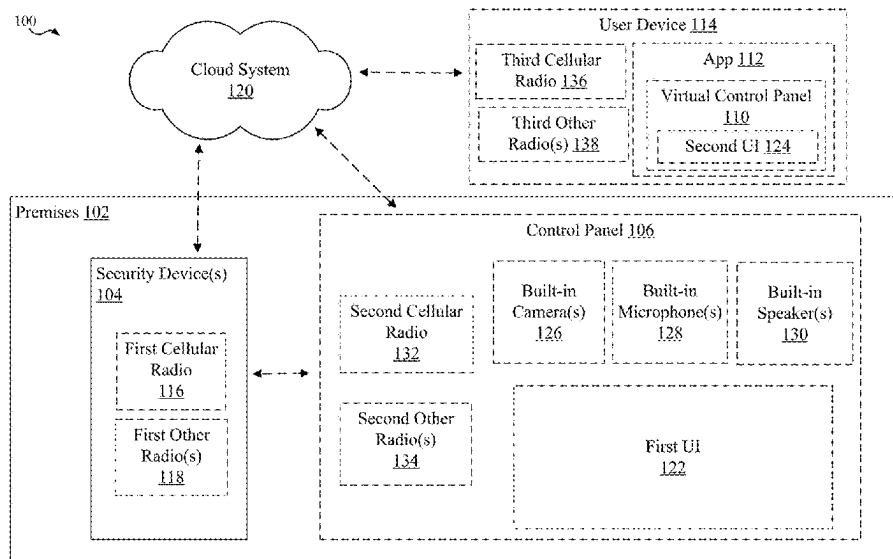
Primary Examiner — Eugene Yun

(74) *Attorney, Agent, or Firm* — ArentFox Schiff LLP

(57) **ABSTRACT**

Example implementations include a method, apparatus, and computer-readable medium comprising receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device; determining, by the control panel, whether the device is a known device; and automatically disarming the control panel responsive to recognizing the device as the known device.

20 Claims, 7 Drawing Sheets



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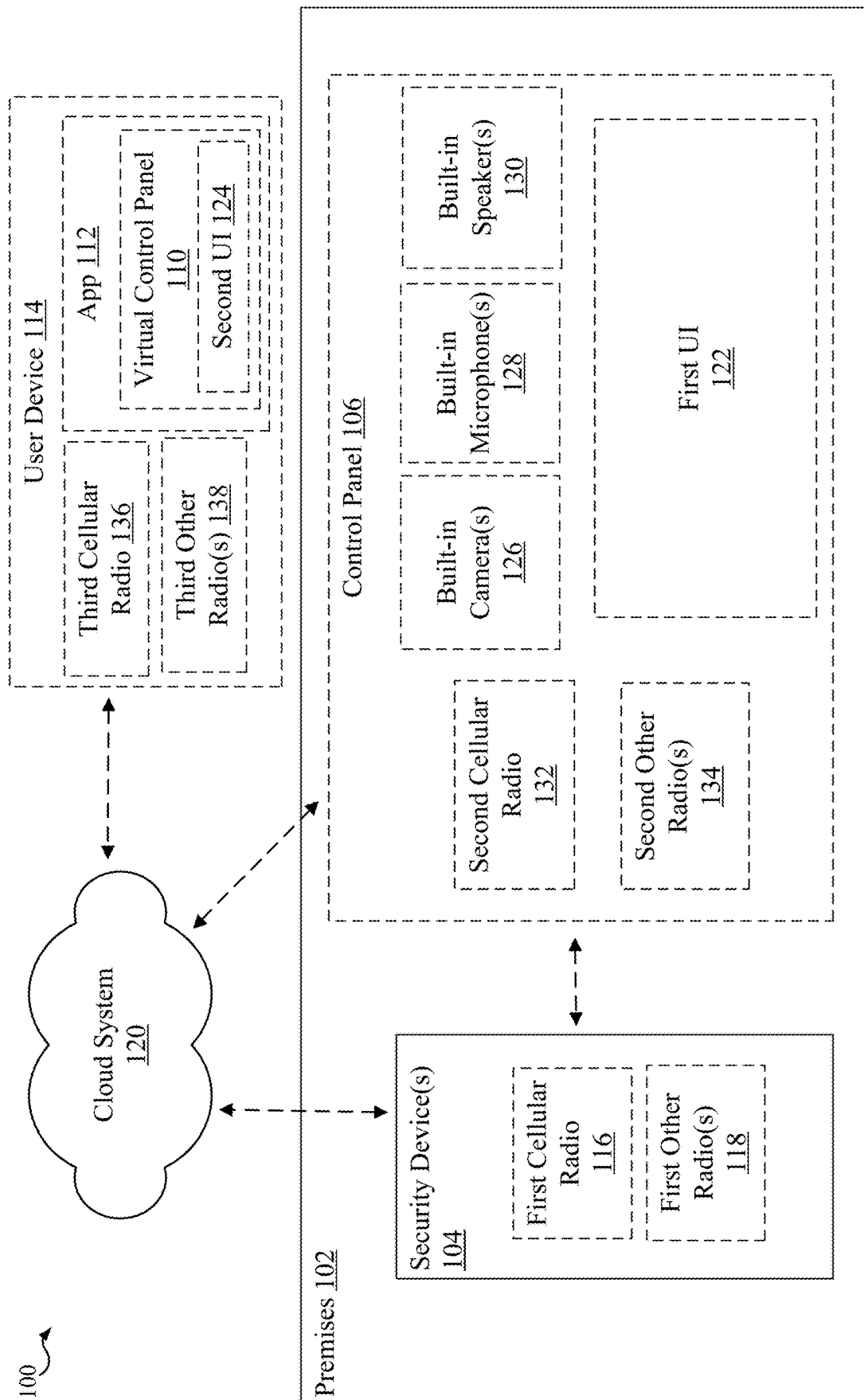
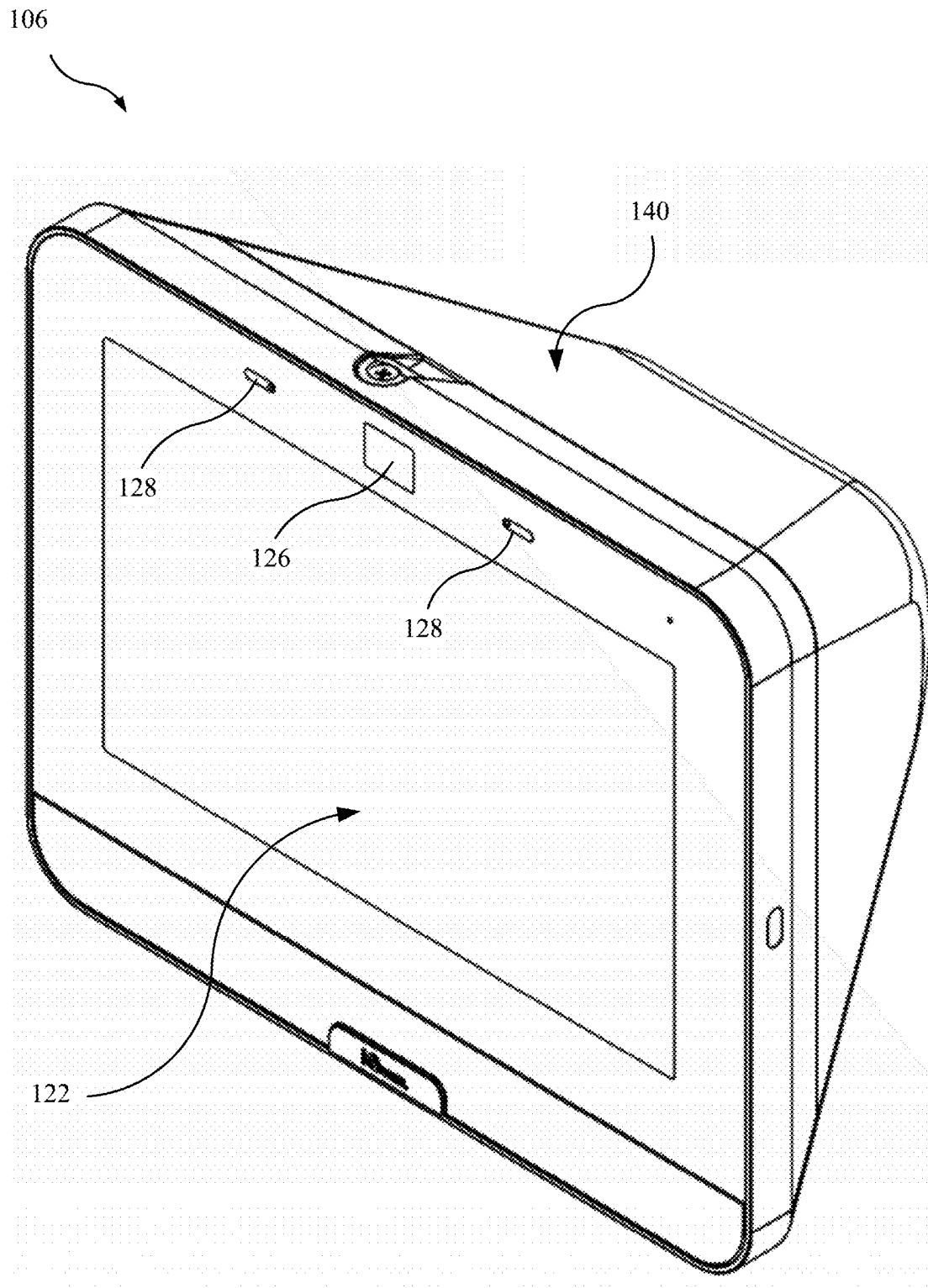


FIG. 1

**FIG. 2**

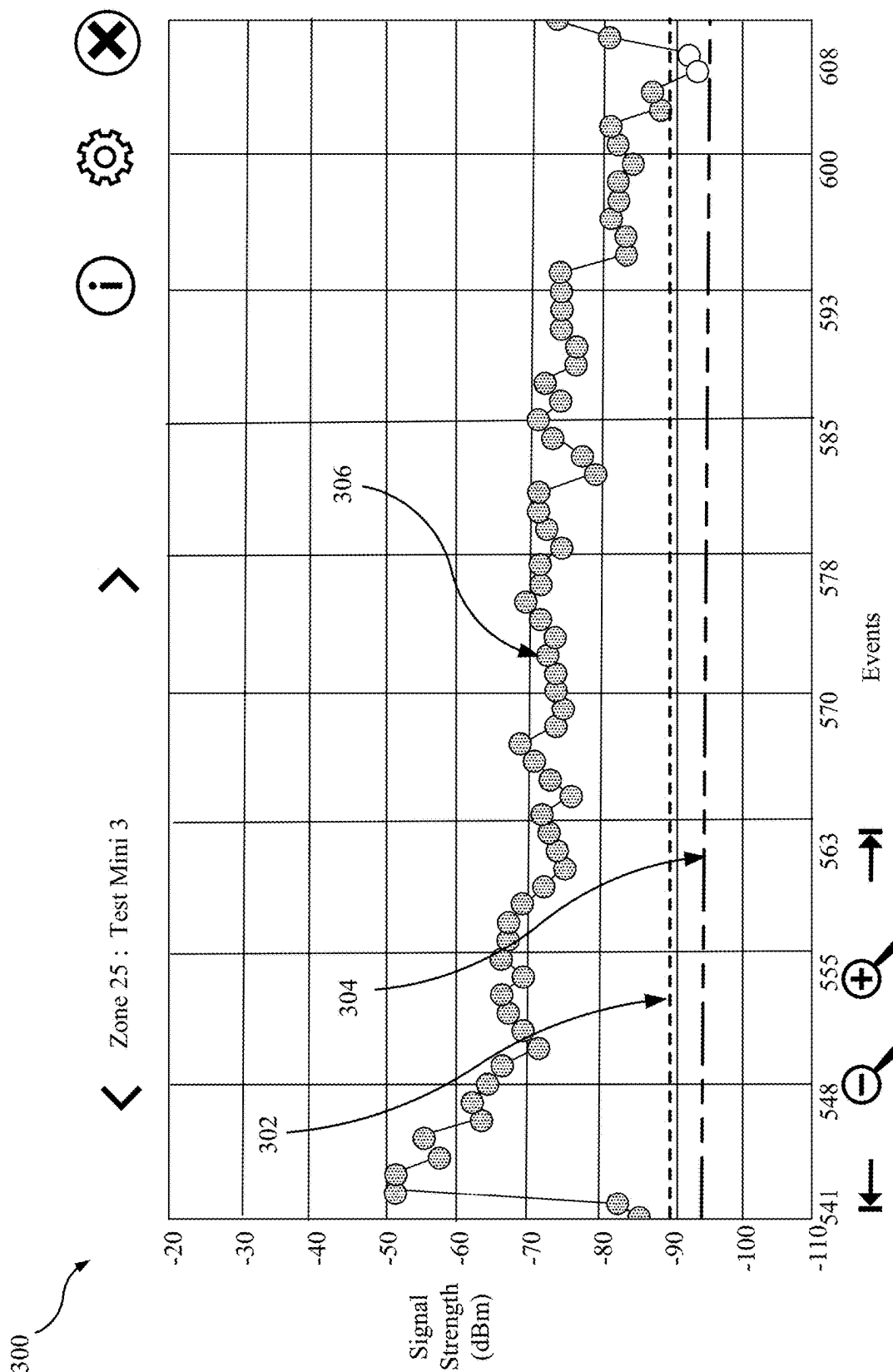


FIG. 3

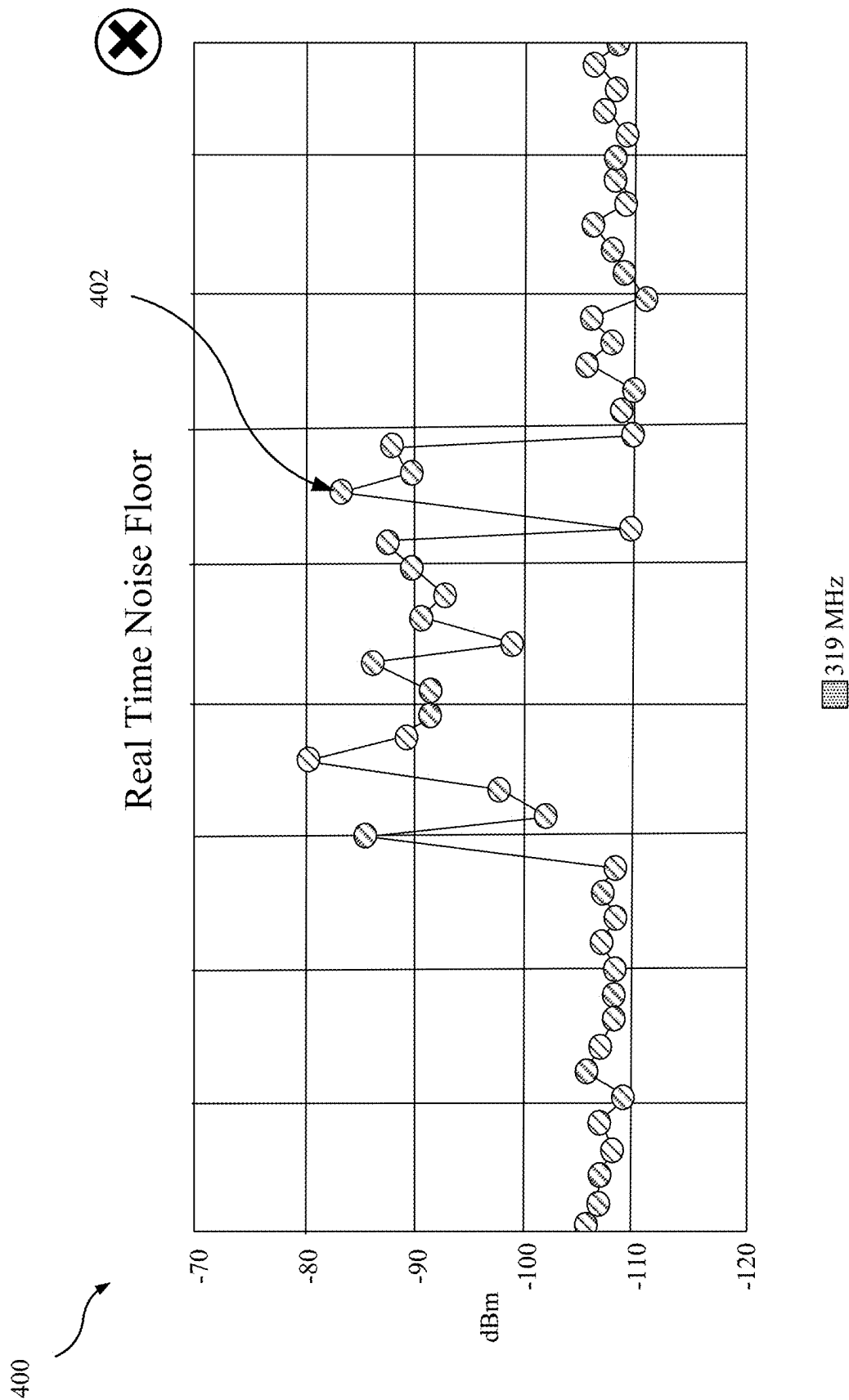
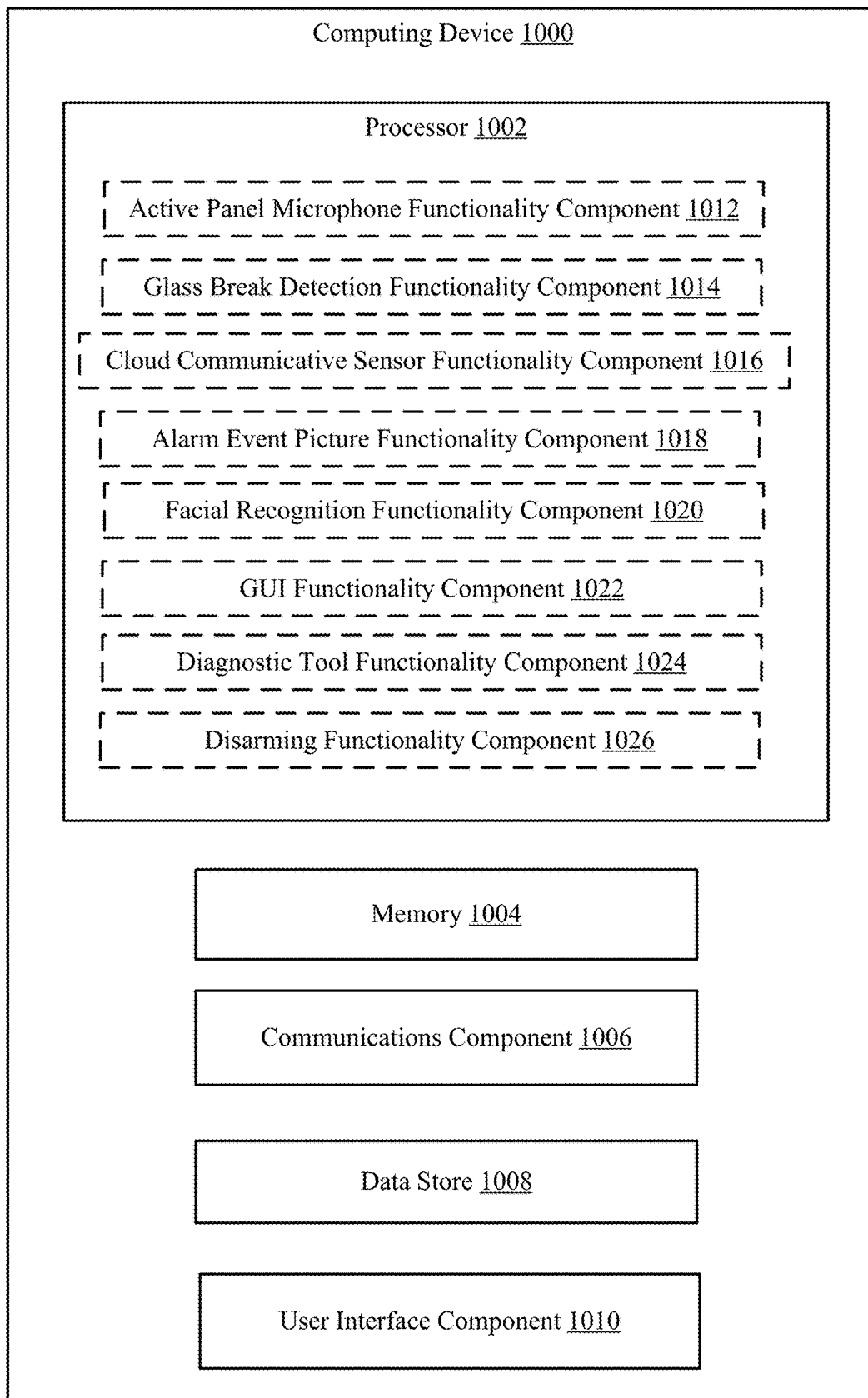
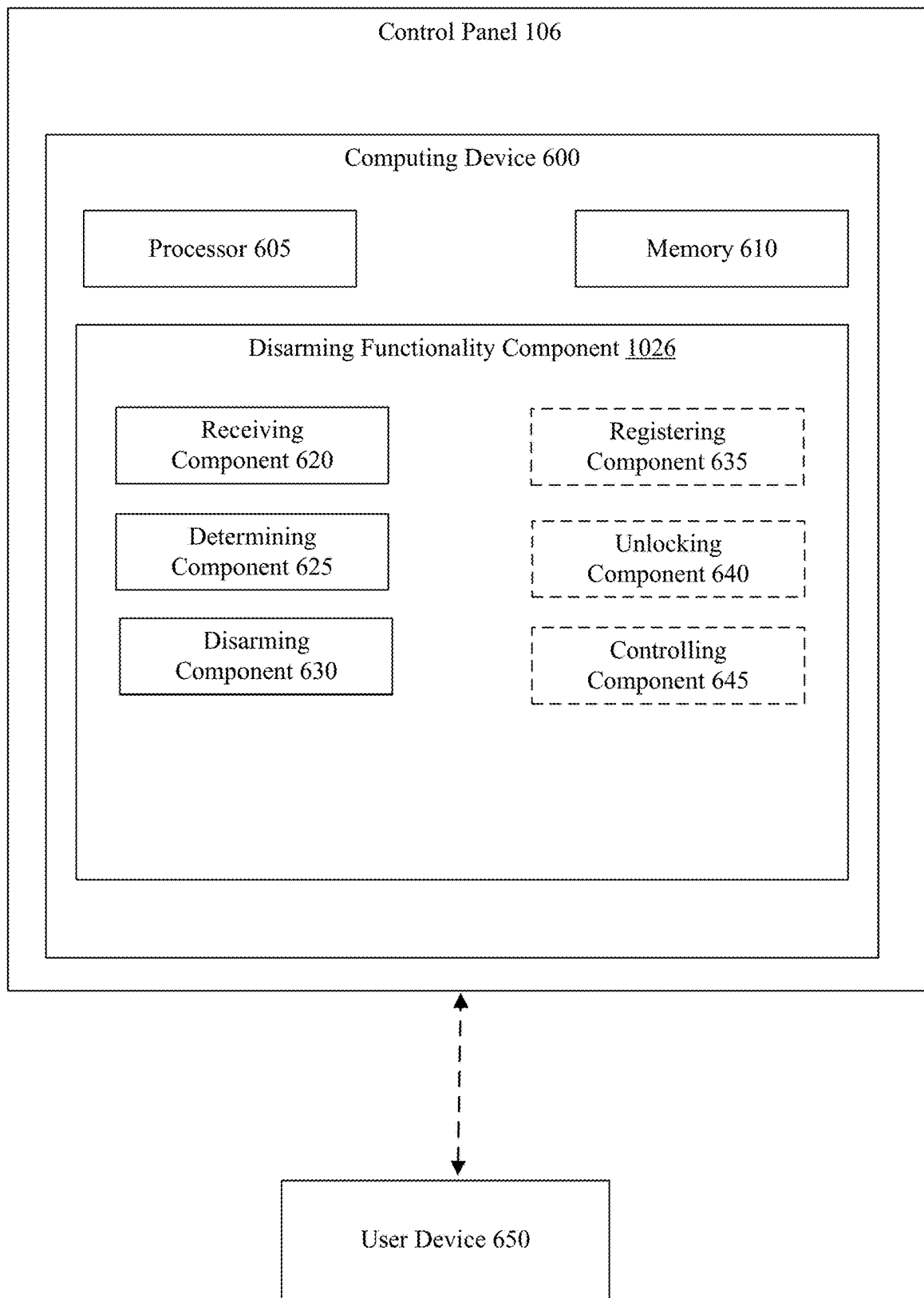
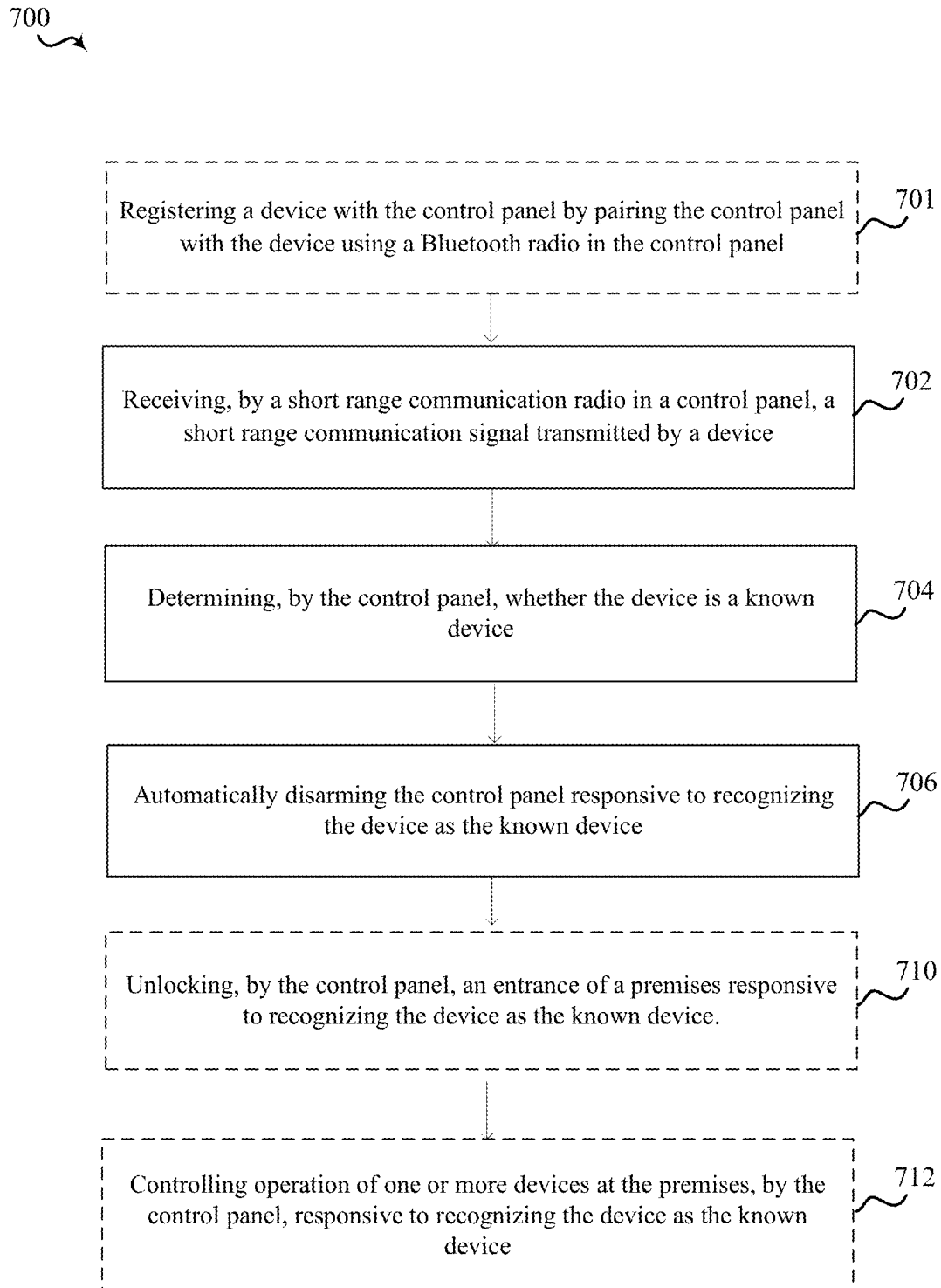


FIG. 4

**FIG. 5**

**FIG. 6**

**FIG. 7**

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SECURITY / AUTOMATION SYSTEM CONTROL PANEL WITH SHORT RANGE COMMUNICATION DISARMING

CROSS REFERENCE TO RELATED APPLICATION(S)

This application is a continuation of U.S. non-provisional application Ser. No. 17/412,054, entitled "SECURITY/AUTOMATION SYSTEM CONTROL PANEL WITH SHORT RANGE COMMUNICATION DISARMING" and filed on Aug. 25, 2021, which claims the benefit of U.S. Provisional Application Ser. No. 63/151,363, entitled "CLOUD SECURITY/AUTOMATION SYSTEM" and filed on Feb. 19, 2021, both of which are expressly incorporated by reference herein in the entirety.

FIELD

The present disclosure relates generally to security/automation systems and methods.

SUMMARY

The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

An example implementation includes a method comprising receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device. The method further comprises determining, by the control panel, whether the device is a known device. The method further comprises automatically disarming the control panel responsive to recognizing the device as the known device.

Another example implementation includes an apparatus comprising a memory and a processor communicatively coupled with the memory. The processor is configured to receive, by a short range communication radio in a control panel, a short range communication signal transmitted by a device. The processor is further configured to determine, by the control panel, whether the device is a known device. The processor is further configured to automatically disarm the control panel responsive to recognizing the device as the known device.

Another example implementation includes an apparatus comprising means for receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device. The apparatus further comprises means for determining, by the control panel, whether the device is a known device. The apparatus further comprises means for automatically disarming the control panel responsive to recognizing the device as the known device.

Another example implementation includes a computer-readable medium storing instructions executable by a processor that, when executed, cause the processor to receive, by a short range communication radio in a control panel, a short range communication signal transmitted by a device. The instructions, when executed, further cause the processor to determine, by the control panel, whether the device is a

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known device. The instructions, when executed, further cause the processor to automatically disarm the control panel responsive to recognizing the device as the known device.

To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosed aspects will hereinafter be described in conjunction with the appended drawings, provided to illustrate and not to limit the disclosed aspects, wherein like designations denote like elements, and in which:

FIG. 1 is a schematic diagram of an example security/automation system, according to some aspects;

FIG. 2 is an isometric view of an example of a control panel in FIG. 1, according to some aspects;

FIG. 3 is a first example graphical user interface (GUI) of a diagnostic tool in a control panel in FIG. 1, according to some aspects;

FIG. 4 is a second example GUI of a diagnostic tool in a control panel in FIG. 1, according to some aspects;

FIG. 5 is a block diagram of an example computing device which may implement all or a portion of any component in FIG. 1, according to some aspects;

FIG. 6 is a block diagram of example components of a computing device which may implement all or a portion of a control panel in FIG. 1 configured for short range communication disarming, according to some aspects; and

FIG. 7 is a flow diagram of an example method for short range communication disarming, according to some aspects.

DETAILED DESCRIPTION

The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known components may be shown in block diagram form in order to avoid obscuring such concepts.

Deconstructed Security/Automation System

Some security/automation systems provide an "All-in-One" control panel that includes hardware features, computing resources, software resources for implementing application intelligence, a user interface (UI), one or more radios, and external communication (e.g., with a monitoring station, a cloud system, etc.). For example, a control panel may include a user interface (e.g., processor and software resources), one or more radios (configured according to a protocol such as, e.g., PowerG, Z-wave, etc.) to wirelessly communicate with associated sensors and automation devices, interfaces to connect to wired sensors, application intelligence (e.g., processor and software resources), and

communication of state to a remote application (according to a protocol such as, e.g., wireless fidelity (Wi-Fi), long term evolution (LTE), etc.).

Generally, Internet of things (IoT) is driving the proliferation of LTE category M (cat-M) and narrowband IoT (NB-IoT) devices (e.g., cheap, low power, cellular connected sensors). Deployment of LTE and Wi-Fi is becoming near ubiquitous, while inexpensive silicon for modern modulation schemes is allowing for improved performance and features for sensors. Some systems provide Wi-Fi with multiple bands, multiple-input multiple-output (MIMO) communication, mesh networking, and cheap Wi-Fi connected cameras. Moreover, computing resources are becoming available in the form of cloud computing (e.g., a private or public cloud system that provides computing and storage resources via access over a network, e.g., Amazon Web Services (AWS)), software as a service (SaaS), on-demand computing for artificial intelligence (AI) and neural networking (e.g., user independent voice recognition, facial recognition), etc. Cell phones and “smart” home assistant devices have also introduced user interaction in the form of voice UIs (e.g., Amazon Alexa, Google, Siri, etc.).

Given the above, some aspects provide a “deconstructed” security/automation system. The user interface of the deconstructed security/automation system may be provided as an application or “app” on a user device (e.g., on a user phone, tablet, computer, bring your own device (BYOD), etc.). The application intelligence of the deconstructed security/automation system may be moved to the cloud, where each customer has a virtual instance of the intelligence, and the instance runs in the cloud and communicates to the UI of a user device wherever the user is and on whatever device the user is using at a given time. The state of the deconstructed security/automation system may be communicated to a remote application (e.g., via Wi-Fi, LTE, etc.). In an aspect, the sensors of the deconstructed security/automation system and their associated radios provide reliable, 2-way, encrypted communication, and the sensors are low power and have long battery life.

In an aspect, the hardware of the deconstructed security/automation system may be configured as a box which may be located in a closet or mounted on a wall (e.g., at a garage). The box may include a router with Wi-Fi MIMO, LTE, sensor radios, and Z-wave, and may be configured for improved antenna performance. The box may have a wide area network (WAN) port to plug into a cable or digital subscriber line (DSL) router. In an aspect, the UI of the deconstructed security/automation system may be provided by an app on a user device (e.g., a phone, a tablet, etc.). The intelligence of the deconstructed security/automation system (e.g., functionality for maintaining state, deciding on actions based on state changes, etc.), voice recognition, facial recognition, etc. may be implemented in the cloud.

In an aspect, the deconstructed security/automation system provides Wi-Fi MIMO, mesh, and real router performance. Accordingly, for example, the deconstructed security/automation system may provide whole home coverage, where mesh nodes are added as needed. The deconstructed security/automation system may also support Wi-Fi cameras with high resolution and high frame rate. The deconstructed security/automation system may allow for integration with other smart devices. For example, in an aspect, the deconstructed security/automation system may allow for integration with a smart television (TV) with an app that shows sensor changes and camera views in a pop-up window while watching TV.

In an aspect, the deconstructed security/automation system implements cloud computing and storage. Accordingly, the deconstructed security/automation system may provide virtually unlimited compute power that may be scaled up or down on demand. In this aspect, the deconstructed security/automation system may allow for voice recognition and/or facial recognition as seamless features that are available from any device with a microphone/camera. In this aspect, software updates to a user’s virtual instance may be flexibly scheduled/performed in the cloud as needed (unlike conventional security/automation systems where updates are performed by a dealer). Various features of the deconstructed security/automation system may be readily turned on/off and billed for. This aspect may also provide cloud storage of images and videos from cameras associated with the system.

In an aspect, the manufacturer or dealer for the deconstructed security/automation system may own the cellular contract with the customer. However, for customers that desire monitored security, state information may go from the cloud of the deconstructed security/automation system to the servers or cloud of the company providing the monitoring service.

By pushing the intelligence to the cloud, the deconstructed security/automation system may provide a “home security/automation system” that is distributed and virtual. The deconstructed security/automation system is no longer limited to a single system and the sensors that are within radio range. Instead, the deconstructed security/automation system may include an aggregate of devices that are associated with an instance of intelligence running in the cloud. As long as a device can communicate to the cloud, the device may be a part of the security/automation system. For example, the system may include IoT devices with LTE cat-M or NB-IoT radios, and the IoT devices may be geographically located anywhere (e.g., the sensors in the system do not need to be within radio range of a control panel). In an aspect, for example, multiple physical installations may be integrated into a single instance for monitoring and control. Alternatively, the system may provide one physical installation for a multi-unit building, and may then provide a separate virtual instance for each unit (e.g., provide partitions).

In an aspect, the system may include a fully integrated control panel. In an aspect, the panel may include a color liquid crystal display (LCD) touchscreen interface that provides an intuitive graphical user interface (GUI) that allows for gesture-based user interaction (e.g., touch, swipe, etc.). In one non-limiting aspect, the panel may include a multi-core processor (e.g., four processor cores) that, while waiting for sensor state changes in the security/automation system, provides additional functionality as described with reference to various aspects herein (e.g., active panel microphones below). In one non-limiting aspect, the panel may include a chipset (e.g., a Qualcomm Snapdragon chipset) that is configured to connect to the Internet via a Wi-Fi and/or cellular network. The chipset may include multiple radios for communication with premises security sensors/devices and/or premises automation sensors/devices. For example, in an aspect, the chipset may include radios for Bluetooth, PowerG, Z-Wave, etc. In an aspect, the sensors/devices of the security/automation system may be wireless and may include, for example, one or more door/window sensors, motion sensors, carbon monoxide detectors, smoke detectors, flood sensors, etc.

In one non-limiting aspect, for example, since the panel can connect to the Internet via a Wi-Fi network or a cellular network, an app may run on a user smartphone or other

mobile device (e.g., a tablet, a wearable, etc.). The user may use the app to remotely control various features of a premises security/automation system, for example, by a gesture on a user interface of the app (e.g., by a touch, swipe, etc.), or view images/video from a camera. For example, the user, who may be remote from a premises and who is planning to return to the premises, may use the app to remotely turn a porch light on or to remotely change a setting on a heating, ventilation, and air conditioning (HVAC) thermostat, so that the premises is comfortable when the user arrives at the premises.

Active Panel Microphones

In one non-limiting aspect, the panel may include one or more microphones that can be utilized to monitor the ambient noise in a protected area (e.g., a premises). In one aspect, for example, the panel may include one or more software, hardware, and/or firmware modules that implement AI algorithms to recognize normal household voices and activity patterns. The user may put the panel into a monitoring mode where the panel sends an alert if the panel hears: (a) any voices in the protected area at a time when there typically is none, such as the middle of the night; (b) unknown voices in the protected area at a time when there typically is none, such as the middle of the night; (c) any unknown voices regardless of the time of day or activity period. Using this data, a user may configure the panel to either initiate an alarm or simply notify the user. In another aspect, voice activation/commands along with AI algorithms can be used to configure and use the panel. Accordingly, the panel may include built-in processing power (e.g., the digital signal processing (DSP) implemented by a processor of a chipset in the panel, such as the Qualcomm Snapdragon) and built-in sensors/microphones to implement ambient noise-related event detection, without requiring a separate sensor/device to be installed at a premises.

In an aspect, when the panel is triggered by any of the above conditions, the panel may send a corresponding notification, for example, to a mobile app through a cloud system. In an aspect, when the panel is triggered, the panel may also use a built-in camera to take still images or a video clip and send the images or the video clip to the cloud system, which may then send the images or the video clip to a mobile app or web app on a user device (e.g., a smartphone) for visual verification of an event that triggered the panel. In one non-limiting aspect, for example, AI algorithms in the panel or in the cloud are modeled to scan for unidentified persons, smoke, or other events in the video clip for visual verification. In one non-limiting aspect, for example, a video clip that includes fifteen seconds before and fifteen seconds after the actual event is sent as notification to the cloud.

In an aspect, the panel may be configured to detect events based on various noise detection models, such as continued noise level above a threshold, noise associated with multiple short sharp impacts (e.g., an intruder trying to kick down a door), gunshot detection, voice recognition to identify a request for assistance (e.g., a person falling down and asking for help), glass break detection, or detection of a particular standardized pattern of beeps such as the temporal-three pattern of a smoke detector going off (according to International Organization for Standardization (ISO) 8201 and American National Standards Institute (ANSI)/American Standards Association (ASA) S3.41 Temporal Pattern), the temporal-four pattern of a carbon monoxide detector going off, etc.

For example, in an aspect, the panel may use one or more built-in microphones to detect a fire event based on detecting

the temporal-three pattern of a smoke detector alarm and/or the temporal-four pattern of a carbon monoxide detector alarm. Accordingly, the panel may implement fire detection functionality without requiring a wired or wireless connection with any fire detection sensors such as smoke detectors or carbon monoxide detectors. In one non-limiting aspect, the panel may voice announce fire or CO based on detecting these patterns.

In another aspect, for example, the panel may be configured to use one or more built-in microphones to perform occupancy detection (e.g., for senior care). For example, the panel may use the built-in microphones to detect the ambient noise at a premises and analyze the ambient noise to determine activity of a senior (e.g., whether the senior got out of bed, operated a kitchen appliance, watched TV, etc.). The panel may report such activity of the senior to a remote user (e.g., to a relative of the senior) via an app on a smartphone of the user. In one non-limiting aspect, for example, voice commands can be given to the panel to activate emergency services.

In an aspect, the panel uses built-in processing resources to implement AI algorithms for analyzing various discrete events and for determining what to do in response to a single detected event or in response to multiple detected events. Accordingly, an event may be a triggering point for taking certain actions. In an aspect, for example, the AI algorithms may be downloaded to the panel from a server and may be customized for each individual panel.

In an aspect, the panel may allow for integration of multiple events. For example, the panel may detect multiple unrelated events, and then correlate/infer an integrated event from the multiple unrelated events using built-in AI algorithms. For example, the panel may detect multiple front door open/close events reported by a door contact switch, while a Bluetooth radio of the panel may also detect multiple unrecognized devices/smartphones within range at the premises, and/or the panel may detect an unrecognized person by the AI algorithms running on imagery captured by the internal panel camera and/or by external cameras. The panel may then infer that a gathering is happening at the premises.

In an aspect, the built-in microphone of the panel may continuously listen and may sample the ambient noise at regular intervals to detect audio events, and at the same time the panel may receive reports of other events via various built-in radios such as a Bluetooth radio. In this aspect, the panel has intelligence to correlate multiple concurrently happening events based on an AI model. The AI model may change depending on how a user intends to correlate various concurrently happening events, for example, based on a certain anomaly or a use case desired by the user. For example, the AI model may be configured to take no action when a glass break event is detected while no other event is concurrently detected, but generate an alarm when a glass break event is detected concurrently with another event. In an aspect, the AI modeling and anomaly detection may be dynamically implemented and changed.

In an aspect, the panel may use built-in processing power and one or more built-in microphones to virtually create and simulate one or more sensors. For example, the panel may use one or more built-in microphones and added application to virtually create a fire detection sensor as described above (e.g., by detecting audio patterns of a smoke detector going off) or to virtually create a glass break detection sensor as described below. In an aspect, such virtually created and

simulated sensors may either replace or augment respective dedicated physical sensors in a security/automation system of a premises.

In an aspect, the panel itself may also be virtualized. In an aspect, for example, the panel may use built-in microphones/sensors to virtualize and integrate various simulated sensors to take input in, and then the processing and intelligence applied to the input may be performed in a cloud system in communication with the panel.

Acoustic Signature Detection

In some aspects, the panel may use one or more built-in microphones to detect an acoustic signature associated with one or more events. For example, the panel may include one or more built-in microphones that can be utilized to monitor the ambient noise in a protected area and determine whether the ambient noise includes an acoustic signature associated with an event. In some aspects, for example, the panel may receive sound waves and compare them to one or more of a plurality of known acoustic signatures associated with one or more events such as: a glass break, a gunshot, a dog barking, a person shouting, a smoke detector alarm, a voice, one or more keywords, or any other number of configurable sound events.

In one non-limiting aspect, for example, the panel may perform glass break detection using one or more microphones. For example, the panel may include one or more built-in microphones that can be utilized to monitor the ambient noises in a protected area to detect a glass break event.

In an aspect, the panel may go into a low-power sleep mode, and may then wake up upon detecting a first sound from a probable glass break. After waking up, the panel may continue to analyze subsequent noises detected by the one or more microphone to determine if an actual glass break has occurred.

A glass break event generates a sound with a particular acoustic signature which starts with a thump sound and then follows with a crashing noise. Accordingly, the panel may execute an application that, using the microphones in the panel, is configured to detect a glass break event by identifying a sequence of sounds corresponding to the acoustic signature of a glass break event. For example, in an aspect, the panel has built-in processing power to execute software code to continually listen to the built-in microphones of the panel to detect a thump sound, and may then continue listening to the built-in microphones to determine if a crashing noise associated with a glass break event follows the thump sound. Accordingly, a control panel at a premises may include built-in processing power and built-in sensors/microphones to implement glass break detection functionality without requiring a separate glass break detection sensor/device to be installed at the premises.

Cloud Communicative Sensors

In some security/automation systems, the sensors are short range devices that talk directly to a control panel using wired or wireless connections. However, in an aspect, a security/automation system includes sensors that talk directly to a cloud system, rather than going to the panel first. In an aspect, each sensor device may have a built-in cellular radio, so that the sensor device may use the cellular network to send information directly to a dedicated cloud. Such cloud communicative sensors remove the requirement for the panel to be a physical unit within a protected area. In other words, the panel may be a cloud-based application accessible on a fixed or mobile device that can be located and controlled at any geographic location. The cloud communicative sensors also allow the panel to become increas-

ingly complex as the panel is no longer bound by physical hardware, software, or memory constraints. As technology improves, the panel application may also improve seamlessly.

In some aspect, one or more sensors may use a cellular radio to communicate with a cloud system that supports a security/automation system. In an aspect, one or more sensors may each include a radio configured for communication according to the NB-IoT protocol. The NB-IoT protocol is designed and configured at hardware and at protocol level for small widely-deployed battery-powered devices that only need to communicate infrequently, such as a water meter that connects and reports on a daily basis. In an aspect, for example, an NB-IoT radio may be included in a contact or PIR motion sensor (e.g., a door/window sensor, motion detector, etc.) such that the sensor may connect to a cellular network to send events and other information directly to the cloud.

In an aspect, a security/automation system may include a virtualized control panel and may provide state management and intelligence in a dedicated cloud that can be hosted in a private or public cluster (e.g., AWS, private data center, etc.). Accordingly, any devices that are capable of establishing a direct cellular connection with the cloud may be configured as a part of the security/automation system, such as one or more NB-IoT sensors configured to communicate directly with the cloud using a cellular connection. In an aspect, the NB-IoT sensors of such a security/automation system may be located at various different geographic locations. For example, in one non-limiting aspect, a security system may include one or more cameras that use a cellular radio to send video clips to the cloud when the local AI algorithms detect unidentifiable persons or objects.

In an aspect, instead of configuring the security/automation system via a physical control panel, a user may use a virtual control panel provided by a mobile app that is configured as an interface to the cloud. For example, the user may use a controlling application (app) on a user device to connect to the cloud and configure the security/automation system, e.g., manage and monitor sensors (e.g., turn sensors on or off), implement new sensors in the security/automation system, remove one or more sensors from the security/automation system, etc.

In one non-limiting aspect, for example, such a virtualized control panel may allow for aggregating the security/automation system of multiple buildings together. For example, in an aspect, a user may own two properties at two different physical locations, and may use a single virtualized control panel to monitor both locations.

In an aspect, the virtualized control panel may allow for establishing a hierarchical security/automation system that includes several buildings. For example, at a highest hierarchical level, the virtualized control panel may be configured to indicate whether there are any issues reported at any of the geographical locations of buildings in a geographically distributed security/automation system, while a lower hierarchical level may provide more granularity and further details of issues reported to the security/automation system, such as a state, a city, a specific building, or a specific room where an issue was detected and reported.

In an aspect, the virtualized control panel may allow for configuring a security/automation system that blankets a region. In an aspect, the virtualized control panel may allow for configuring a security/automation system that blankets the assets of a business. In an aspect, for example, the virtualized control panel may allow for configuring a security/automation system that includes a number of NB-IoT

sensors installed at various geographically distributed public utility structures. In one non-limiting aspect, for example, the virtualized control panel may allow for configuring a security/automation system that includes one or more door/window contacts, and/or cellular cameras at the entrance kiosk of state parks, national grid substations, high voltage transmission towers, and/or other national infrastructures.

In another non-limiting aspect, for example, the virtualized control panel may allow for configuring a security/automation system that includes a contact sensor at a mailbox, where the contact sensor communicates directly to the cloud to indicate at what times the mailbox has been opened. Accordingly, the security/automation system may send a notification to a user if the mailbox has been opened/accessed at an odd hour (e.g., between midnight and 5:00 am).

Alarm Event Pictures

In one non-limiting aspect, a control panel may include a built-in forward facing camera. In an aspect, the camera may be used to take a picture of the person who interacts with the panel to arm or disarm the panel and/or set-up the security/automation system and/or the panel. In an alternative or additional aspect, the camera may be used as a motion detector. In an aspect, for example, the panel may delay taking alarm event pictures until motion is detected (e.g., by the panel or by a sensor in communication with the panel) or the local AI algorithm detects an unrecognized person. Accordingly, the panel may not waste memory storage space on meaningless pictures. For example, in an aspect, the panel may detect an alarm event and trigger a siren and/or alert a monitoring center/homeowner. At the same time, the panel may wait until motion is sensed/detected (e.g., by the panel or by a sensor in communication with the panel). Only after motion is sensed/detected, the panel may begin recording video or taking pictures to assist with the determination of who or what caused the alarm event. By waiting until motion is detected or the local AI algorithm detects an unrecognized person, the panel avoids taking unnecessary pictures and therefore retains more memory for pictures that have a greater likelihood of being material to the alarm event.

In one non-limiting aspect, the panel performs motion detection by comparing subsequent frames captured by a built-in camera in the panel. In an aspect, for example, if a door is opened while the panel is in an armed state, the built-in camera continuously captures images and/or video, and the panel performs frame-by-frame comparison of the images and/or video captured by the built-in camera to detect motion based on the amount of change in the pixels of subsequent frames. In one non-limiting aspect, for example, in order to detect motion, an optimized algorithm selectively samples for pixel changes in a frame. The algorithm may be calibrated to ignore pets and other unwanted objects. After motion is detected, the panel starts recording the images/video captured by the built-in camera and sends the recorded images/video to the cloud. The cloud may then send the recorded images/video to a device of a user (e.g., a smartphone, a tablet, etc.) for viewing on an app running on the device of the user.

In one non-limiting aspect, for example, when a person disarms the panel, a user may be notified via an app on the user smartphone that the panel has been disarmed. The user may then use the app to remotely view an image or video of the person who disarmed the panel, where the image or video is taken by a built-in camera in the panel at the time the panel was disarmed or immediately after the panel was disarmed.

In another non-limiting alternative or additional aspect, the user may use the app to remotely view images and videos of the premises taken by a built-in camera of the panel. In response to determining that a service person has arrived at the premises, the user may use the app to remotely disarm the panel.

Facial Recognition by the Control Panel

In one non-limiting aspect, the control panel may include a built-in camera and may use the built-in camera to implement facial recognition. In an aspect, for example, when a person is arming or disarming the panel, the panel may use the built-in camera to take video and/or images of the person and perform facial recognition based on the captured video and/or images to identify the person and determine whether the person is legitimate and authorized to arm or disarm the panel. In an aspect, the panel may use facial recognition in addition to another form of authentication (e.g., passcode, voice recognition, etc.) to perform multi-factor authentication and determine whether the person is legitimate and authorized to arm or disarm the panel.

In an aspect, upon recognizing the person, the panel may control one or more devices to operate according to a desired setting of the recognized person. For example, the panel may turn some lights on or off, turn music or radio on or off, adjust an HVAC temperature setting to a desired temperature, etc.

In another aspect, for example, when the panel is next to a premises entry point such as a door, and a door contact sensor indicates to the panel that the door has been opened, the panel may use the built-in camera to take images of the person passing by and perform facial recognition, optionally together with voice recognition or other sensors, to determine whether the person is legitimate and authorized to enter the premises.

In an aspect, the panel may use facial recognition, optionally together with voice recognition or other sensors, to determine how many people are present at a premises and whether known or unknown people are present at the premises. In one non-limiting aspect, for example, the panel may identify, via a built-in Bluetooth radio, that a number of Bluetooth devices are in range, which indicates a possibility of multiple people being present at the premises. The panel may then use facial recognition (via a built-in security camera), and optionally together with voice recognition (via a built-in microphone) to determine how many people are present at the premises and whether any of those people are legitimate and authorized to be at the premises.

In an aspect, the panel may use a combination of the above to determine whether an unusual event is happening at the premises. For example, the panel may determine whether a number of unrecognized faces have passed by, whether a door has been opened and closed an unusually large number of times, whether an unusually large number of Bluetooth devices are in range, whether a noise sensor is indicating an unusually high amount of noise, whether an infra-red (IR) sensor is detecting an unusually large number of bodies, etc.

In an aspect, the panel may use facial recognition for generating an alarm. For example, the panel may initiate an alarm upon recognizing one or more specific individuals.

GUI Functionality

In one non-limiting aspect, the control panel may implement AI functionality for tracking the applications and functions that a particular user typically invokes and/or is allowed to access. Accordingly, when the panel recognizes a person (e.g., through voice or facial recognition), the panel

may bring up and display GUI features (e.g., buttons, icons, apps, etc.) that are typically invoked by and/or associated with the recognized person.

In an alternative or additional aspect, the panel may allow for restricting one or more features for one or more recognized user. For example, the panel may allow for implementing parental control to limit access to certain features that are otherwise controllable via the panel.

In an aspect, the panel may bring up personalized GUI features of a specific person based on facial recognition using a built-in camera in the panel, as described above.

In an aspect, the AI algorithms for facial recognition are executed by a built-in multicore processor of the panel. In an alternative or additional aspect, the panel may send the images/video captured by the built-in camera to the cloud, and the AI algorithms for facial recognition are executed in the cloud. The cloud then sends the outcome of the facial recognition back to the panel. For example, if the cloud recognizes a person by applying facial recognition to images/video captured by the built-in camera of the panel, the cloud may send the identity of the recognized person to the panel.

Window/Door Sensors Using Capacitive Sensing

In one non-limiting aspect, a security/automation system may include a window/door sensor that implements capacitive sensing to detect if a door or window is open or closed. Such sensors are beneficial because the sensors do not use the conventional magnetic reed switches or other mechanical designs that require two separate pieces to be installed.

Specifically, for example, some sensors used in security/automation systems for detecting whether a window/door is open or closed use a magnet and reed switch. The sensor containing the reed switch, or other magnetic sensing device, is typically mounted on or in the window/door frame, and the magnet is mounted on the window/door. When the window/door is closed, the magnet is in close proximity to the reed switch, keeping it closed. When the window/door is opened, the magnet moves away from the reed switch, causing it to open. The sensor detects the change of state of the reed switch and transmits to a control panel, using wired or wireless communication. These sensors therefore have two components—the sensor device, and the associated magnet.

It is desired to make the window/door sensor as small and inexpensive as possible. As small magnets with high magnetic field strength are a significant part of the overall sensor cost, it is advantageous to eliminate the magnet and use a different method to detect if the window/door is open or closed.

Accordingly, some aspects sense the proximity of the window/door to the window/door frame without a magnet by measuring the capacitance between two conductive measurement points. When the window/door is open, the capacitance will be lower compared to when the window/door is closed and physically close to the two measurement points. A microcontroller with appropriate circuitry may periodically measure the capacitance and then determine whether the window/door is open or closed.

Additionally, in some aspects, the sensing device may have a mode to self-calibrate when it is installed so it knows the difference between open and closed, thus accounting for differences in capacitance caused by different materials (such as wood, metal, masonry, etc.), different physical spacing between the sensor and the window/door, etc.

Additionally, in some aspects, the device may keep a long term history of any drift in values caused (for example) by

changes in the moisture content of a wood window/door, changes in spacing caused by seasonal shifting or settling of construction, painting, etc.

In an aspect, for example, the window/door sensor includes an electrical circuit capable of measuring the capacitance between two closely-spaced metal elements, and the capacitance between the two metal elements changes depending on their proximity to the window/door. The metal elements can be implemented as patterns in the copper plating on a printed circuit board, or as separate metal elements connected to the measurement circuitry.

In an aspect, the window/door sensor may be calibrated/trained during installation by opening and closing the window/door multiple times. In an aspect, for example, in order to calibrate the sensor, the capacitive coupling of the sensor may be measured at different states of the window/door (e.g., fully open, fully closed, half open, etc.).

In an aspect, the sensor threshold settings derived by calibration may vary depending on the material of the window/door (e.g., metal, wood, glass, etc.) and/or depending on the location/orientation of the sensor on the window/door and/or on the frame of the window/door.

Bass Augmentation Sub-System (BASS)

In one non-limiting aspect, a control panel may include multiple primary speakers and a modular back speaker. In an aspect, the panel implements a modular speaker that may be attached or removed and which improves the sound qualities of the panel. The panel may use the modular speaker and one or more microphones to communicate with users through voice commands and responses, and may also use the modular speaker to broadcast other messages and music. Accordingly, the panel may function as a home appliance that communicates clearly and effectively.

In an aspect, the panel may allow for audio as well as video user interaction. For example, the panel may include one or more speakerphones and microphones. In one non-limiting aspect, for example, if the panel goes into an alarm condition, the panel may report an alarm to a monitoring center and may use a cellular interface of the control panel to establish a two-way voice call between the panel and the monitoring center. For example, in an aspect, in response to a reported event, the monitoring center may make a voice call to the panel and ask a homeowner, via one or more speakerphones on the panel, about any emergencies existing at the premises and/or whether the homeowner requires assistance.

Diagnostic Tool

One non-limiting aspect implements a diagnostic tool that tests, measures, and/or graphically maps the signal strength of the connection between one or more sensors and the control panel. Accordingly, a technician may do diagnostic analysis by using the panel itself, rather than needing to use additional signal strength meters, etc. This may speed up the installation of a security/automation system and/or provide a more robust installed security/automation system.

In one non-limiting aspect, the diagnostic tool may measure the received signal strength of wireless signals between one or more sensors and one or more radios in the panel (e.g., cellular or other radios). In an aspect, the diagnostic tool reads the received signal strength indicator (RSSI) of a radio. For example, the diagnostic tool may listen to a radio and determine an instantaneous RSSI related to the background noise and plot the instantaneous RSSI on a graph over time. Accordingly, a technician/installer may use the graph to identify sources of noise in the environment that would interfere with the operation of the security/automation system.

For example, in an aspect, RSSI of sensor radios is sent to the cloud and historic signal strength data is maintained. By observing the historic graphical information over time, a technician/installer may notice that operation of an electrical device is producing a signal that interferes with the signal transmitted by a sensor or camera that is trying to communicate with the panel. The technician may then adjust the installation of the panel, sensor, or camera and/or the electrical device to mitigate the interference.

In an aspect, the graph may also display the average of the background noise by a first horizontal bar, and may also display a minimum acceptable sensor RSSI by a second horizontal bar. Accordingly, the technician/installer may observe the graph over time and discern whether the RSSI of a sensor is above the second bar, thus being acceptable. In an aspect, various data points in the graph may be color-coded to indicate different signal quality categories, e.g., good signal, marginal signal, unreliable signal, etc. In an aspect, for example, when the graph color-codes a data point of a sensor as being marginal or unreliable, the technician/installer may reposition the sensor and/or reposition the panel until subsequent data points of the sensor are color-coded in the graph as being a good signal.

In an aspect, repositioning the sensor may include changing the location and/or changing an orientation of the sensor. Similarly, repositioning the panel may include changing a location and/or changing an orientation of the panel.

Use of a Short Range Communication Radio for Disarming

In one non-limiting aspect, a short range communication radio of the control panel may be used to determine whether a known device is in range, and the control panel may be automatically disarmed in response to the known device being in range. In one non-limiting aspect, an entrance of a premises may also be unlocked in response to the known device being in range. In an aspect, the short range communication radio may be, for example, but is not limited to, a Bluetooth radio, a Bluetooth Low Energy (BLE) radio, a near-field communication (NFC) radio, etc.

In one non-limiting aspect, for example, when a panel recognizes a BLE signal from a known device, the panel is automatically disarmed. In an aspect, instead of arming/disarming the panel by determining the exact location of a user, the panel arms/disarms based on detecting that a user is within BLE range and that a BLE device of the user has been registered with the panel.

In one non-limiting aspect, for example, a user smartphone may be paired with the panel in a premises, e.g., using a built-in Bluetooth radio in the panel. In one non-limiting aspect, for example, when the user approaches a front or back door of the premises, the built-in Bluetooth radio in the panel may detect that the user smartphone is within range. In response to detecting that the user smartphone is within range, the panel may disarm and send a Z-Wave command to unlock the door and/or turn on the lights at the premises. In one non-limiting aspect, if the user opens the back door of the premises, a sensor on the back door may send a signal to the panel to indicate that the back door has been opened. In response, the panel may, for example, chime and play an audio message such as "Back door opened!" In another non-limiting aspect, the built-in Bluetooth radio in the panel may be used to pair new security or home automation sensors with the panel. For example, in an aspect, pairing the sensor may be performed from a mobile phone app by scanning the sensor QR code and sending the sensor details to the panel using Bluetooth.

Turning now to the figures, example aspects are depicted with reference to one or more components described herein, where components in dashed lines may be optional.

Referring to FIG. 1, a security/automation system 100 of a premises 102 may include various security devices 104 (e.g., sensors, cameras, etc.) installed/positioned throughout the premises 102. In some aspects, at least some of the security devices 104 may include a first cellular radio 116 for communicating directly (e.g., via a cellular network) with a cloud system 120 that implements at least some of the functionalities provided by the system 100, as described herein with reference to various aspects. Alternatively and/or additionally, at least some of the security devices 104 may communicate with the cloud system 120 via another wired or wireless connection, for example, via a physical Ethernet connection.

In an aspect, at least some of the security devices 104 may communicate with a control panel 106 that is physically installed at the premises 102. In an aspect, for example, at least some of the security devices 104 may include one or more first other radios 118 (e.g., Bluetooth, PowerG, Z-Wave, Zigbee, etc.) for communicating with the panel 106 that includes one or more corresponding second other radios 134 (e.g., Bluetooth, PowerG, Z-Wave, etc.). Alternatively and/or additionally, at least some of the security devices 104 may communicate with the panel 106 via another wired or wireless connection, for example, via a physical Ethernet connection.

In an aspect, the system 100 may be at least partially configured and/or controlled via a first UI 122 of the panel 106 to implement at least some of the functionalities described herein with reference to various aspects. In an aspect, the panel 106 may include one or more built-in cameras 126, one or more built-in microphones 128, and/or one or more built-in speakers 130 to implement at least some of the functionalities described herein with reference to various aspects.

In some aspects, the panel 106 may include a second cellular radio 132 for communicating directly (e.g., via a cellular network) with the cloud system 120 to implement at least some of the functionalities provided by the system 100, as described herein with reference to various aspects. Alternatively and/or additionally, the panel 106 may communicate with the cloud system 120 via another wired or wireless connection, for example, via a physical Ethernet connection.

In some alternative or additional aspects, the system 100 may be at least partially configured and/or controlled via a second UI 124 of a virtual control panel 110 provided via an app 112 executing on a user device 114 (e.g., a mobile device). In some aspects, the user device 114 may include a third cellular radio 136 for communicating directly (e.g., via a cellular network) with the cloud system 120 to implement at least some of the functionalities provided by the system 100, as described herein with reference to various aspects. Alternatively and/or additionally, the user device 114 may communicate with the cloud system 120 via another wired or wireless connection, for example, via a physical Ethernet connection. In some aspects, the user device 114 may also include one or more other radios 138 (e.g., Bluetooth, PowerG, Z-Wave, etc.).

Referring to FIG. 2, in an aspect, the panel 106 may include a removable back speaker 140 that may also provide support as a stand for placing the panel 106 on a surface such as a countertop or a desk.

FIG. 3 includes a first non-limiting example of a GUI 300 of the diagnostic tool described above with reference to

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some aspects. In FIG. 3, a test/installer tool graph of the signal strength **306** changes from -50 dBm to less than -90 dBm as a sensor is progressively moved away from the control panel **106**. The graph also displays the average of the background noise by a first horizontal bar **304**, and also displays a minimum acceptable sensor RSSI by a second horizontal bar **302** that is above the first horizontal bar **304**.

FIG. 4 includes a second non-limiting example of a GUI **400** of the diagnostic tool described above with reference to some aspects. In FIG. 4, a graph shows a real-time noise floor measurement **402** at the control panel **106**, indicating the interference caused by a switching power supply in an LED light fixture. Specifically, the noise floor is slightly above -110 dBm when the LED light is off, but jumps up to the neighborhood of -90 dBm when the LED light is turned on, and drops down to -110 dBm when the LED light is turned back off again. In other example aspects, alternative or additional sources of noise/interference may include, for example, a microwave oven, a cordless phone, etc.

FIG. 5 illustrates an example block diagram providing details of computing components in a computing device **1000** that may implement all or a portion of one or more components in a control panel, a cloud system, a sensor device, a user device (e.g., a smartphone, a tablet, a laptop computer, a desktop computer, etc.), or any other component described above. The computing device **1000** includes a processor **1002** which may be configured to execute or implement software, hardware, and/or firmware modules that perform any functionality described above with reference to one or more components in a control panel, a cloud system, a sensor device, a user device, or any other component described above. For example, the processor **1002** may be configured to execute an active panel microphone functionality component **1012** to provide active panel microphone functionality, a glass break detection functionality component **1014** to provide glass break detection functionality, a cloud communicative sensor functionality component **1016** to provide cloud communicative sensor functionality, an alarm event picture functionality component **1018** to provide alarm event picture functionality, a facial recognition functionality component **1020** to provide facial recognition functionality, a GUI functionality component **1022** to provide GUI functionality, a diagnostic tool functionality component **1024** to provide diagnostic tool functionality, and/or a disarming functionality component **1026** to provide functionality for using short range communications for disarming, as described herein with reference to various aspects.

The processor **1002** may be a micro-controller and/or may include a single or multiple set of processors or multi-core processors. Moreover, the processor **1002** may be implemented as an integrated processing system and/or a distributed processing system. The computing device **1000** may further include a memory **1004**, such as for storing local versions of applications being executed by the processor **1002**, related instructions, parameters, etc. The memory **1004** may include a type of memory usable by a computer, such as random access memory (RAM), read only memory (ROM), tapes, flash drives, magnetic discs, optical discs, volatile memory, non-volatile memory, and any combination thereof. Additionally, the processor **1002** and the memory **1004** may include and execute an operating system executing on the processor **1002**, one or more applications, display drivers, etc., and/or other components of the computing device **1000**.

Further, the computing device **1000** may include a communications component **1006** that provides for establishing

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and maintaining communications with one or more other devices, parties, entities, etc., utilizing hardware, software, and services. The communications component **1006** may carry communications between components on the computing device **1000**, as well as between the computing device **1000** and external devices, such as devices located across a communications network and/or devices serially or locally connected to the computing device **1000**. For example, the communications component **1006** may include one or more buses, and may further include transmit chain components and receive chain components associated with a wireless or wired transmitter and receiver, respectively, operable for interfacing with external devices.

Additionally, the computing device **1000** may include a data store **1008**, which can be any suitable combination of hardware and/or software, that provides for mass storage of information, databases, and programs. For example, the data store **1008** may be or may include a data repository for applications and/or related parameters not currently being executed by processor **1002**. In addition, the data store **1008** may be a data repository for an operating system, application, display driver, etc., executing on the processor **1002**, and/or one or more other components of the computing device **1000**.

The computing device **1000** may also include a user interface component **1010** operable to receive inputs from a user of the computing device **1000** and further operable to generate outputs for presentation to the user (e.g., via a display interface to a display device). The user interface component **1010** may include one or more input devices, including but not limited to a keyboard, a number pad, a mouse, a touch-sensitive display, a navigation key, a function key, a microphone, a voice recognition component, or any other mechanism capable of receiving an input from a user, or any combination thereof. Further, the user interface component **1010** may include one or more output devices, including but not limited to a display interface, a speaker, a haptic feedback mechanism, a printer, any other mechanism capable of presenting an output to a user, or any combination thereof.

Some further aspects are provided below.

1. A control panel comprising:

one or more microphones;

at least one memory; and

at least one processor coupled with the at least one memory, wherein the at least one processor is configured to:

monitor an ambient noise via the one or more microphones; and

determine whether the ambient noise is indicative of a security event.

2. The control panel of clause 1, wherein, in response to determining that the ambient noise is indicative of the security event, the at least one processor is further configured to initiate an alarm or send a notification to a cloud system.

3. The control panel of clause 1, wherein, in response to determining that the ambient noise is indicative of the security event, the at least one processor is further configured to:

capture one or more still images or videos via a camera of the control panel; and

send the one or more still images or videos to a cloud system, wherein the cloud system is configured to send the one or more still images or videos to a device of a user for viewing on an app running on the device of the user.

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4. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to determine whether the ambient noise includes one or more known or unknown voices. 5
5. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to determine whether the ambient noise is continually above a threshold level for longer than a time period. 10
6. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to determine whether the ambient noise includes a sequence of short sharp impacts. 15
7. The control panel of clause 1, wherein the security event comprises a fire event, wherein, to determine whether the ambient noise is indicative of the fire event, the at least one processor is further configured to determine whether the ambient noise includes a standardized pattern of beeps associated with activation of a fire detection device in response to the fire event. 20
8. The control panel of clause 1, wherein the security event comprises a fire event, wherein, to determine whether the ambient noise is indicative of the fire event, the at least one processor is further configured to determine whether the ambient noise includes at least one of a temporal-three pattern associated with a smoke detector going off or a temporal-four pattern associated with a carbon monoxide detector going off. 30
9. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to perform occupancy detection based on the ambient noise. 35
10. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to: 40
 - determine, based on the ambient noise, whether there is an activity within a monitored area; and
 - send a notification to a cloud system in response to detecting the activity within the monitored area.
11. The control panel of clause 1, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to execute an artificial intelligence module locally at the control panel. 45
12. The control panel of clause 11, wherein the artificial intelligence module is downloaded to the control panel from a server, wherein the artificial intelligence module is customized for the control panel. 50
13. The control panel of clause 11, wherein the artificial intelligence module is configured to determine an action in response to a single detected event or in response to multiple concurrently-detected discrete events. 55
14. The control panel of clause 11, wherein the artificial intelligence module is configured to integrate multiple concurrently-detected discrete events to infer that an integrated event is happening. 60
15. The control panel of clause 14, wherein the multiple concurrently-detected discrete events includes at least two events detected using at least two different sensor types. 65

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16. The control panel of clause 14, wherein the multiple concurrently-detected discrete events includes at least one event detected based on the ambient noise using the one or more microphones.
17. The control panel of clause 14, wherein the multiple concurrently-detected discrete events includes at least one event detected based on a wired or wireless signal received by the control panel from a sensor.
18. The control panel of clause 11, wherein the artificial intelligence module is configured to integrate multiple concurrently-detected discrete events based on one or more user preferences received via the control panel.
19. The control panel of clause 1, wherein the security event is associated with an acoustic signature, wherein, to determine whether the ambient noise is indicative of the security event, the at least one processor is further configured to determine whether the ambient noise includes the acoustic signature.
20. The control panel of clause 19, wherein the security event comprises a glass break event, wherein, to determine whether the ambient noise is indicative of the glass break event, the at least one processor is further configured to:
 - detect a first noise by the one or more built-in microphones, while operating in a low-power sleep mode;
 - wake up from the low-power sleep mode in response to determining that the first noise is associated with a first sound in the acoustic signature of a probable glass break; and
 - analyze subsequent noises to determine whether an actual glass break has occurred, based on subsequent sounds in the acoustic signature.
21. The control panel of clause 19, wherein the first sound comprises a thump sound, wherein the subsequent sounds comprise crashing sounds.
22. A system comprising at least one sensor device that includes:
 - a sensor element configured to sense a physical phenomenon;
 - a cellular radio;
 - a memory; and
 - at least one processor coupled with the memory, wherein the at least one processor is configured to:
 - receive a signal from the sensor element, the signal being indicative of the physical phenomenon as sensed by the sensor;
 - determine whether a level of the signal element is indicative of an alarm condition; and
 - use the cellular radio to send a notification directly to a cloud system in response to the level of the signal being indicative of the alarm condition.
23. The system of clause 22, wherein the cellular radio is operable according to narrowband Internet of things "NB-IoT" protocol.
24. The system of clause 22, wherein the at least one sensor device comprises a fire detection sensor.
25. The system of clause 22, further comprising a virtual control panel implemented as a cloud-based application executable on a user device, wherein the virtual control panel is configured to communicate with the cloud system to interact with or configure the system.
26. The system of clause 25, further comprising at least two sensor devices located at two different geographical locations, wherein each one of the at least two sensor devices is configured to communicate directly with the cloud system, wherein the virtual control panel is configured to communicate with the cloud system to

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configure the at least two sensors or to receive information communicated by the at least two sensors.

27. A control panel comprising:
 at least one camera;
 at least one memory; and
 at least one processor coupled with the at least one memory, wherein the at least one processor is configured to:
 determine that a security event has happened; and
 take one or more still images or videos by the at least one camera subsequent to determining that the security event has happened.

28. The control panel of clause 27, wherein the at least one camera comprises a front-facing camera.

29. The control panel of clause 27, wherein the security event is associated with a motion, wherein, in order to determine that the security event has happened, the at least one processor is further configured to detect the motion using the at least one camera.

30. The control panel of clause 31, wherein, in response to determining that the security event has happened, the at least one processor is further configured to initiate an alarm or send a notification to a cloud system.

31. The control panel of clause 27, wherein the at least one processor is further configured to send the one or more still images or videos to a cloud system, wherein the cloud system is configured to send the one or more still images or videos to a device of a user for viewing on an app running on the device of the user.

32. The control panel of clause 27, wherein the security event is associated with identification or authentication of a user, wherein the at least one processor is further configured to use the one or more still images or videos to perform facial recognition.

33. The control panel of clause 32, wherein the security event comprises a user interaction with the control panel.

34. The control panel of clause 32, wherein the security event comprises arming or disarming the control panel.

35. The control panel of clause 32, wherein the at least one processor is further configured to perform a multi-factor authentication based on the facial recognition and at least one other form of authentication.

36. The control panel of clause 32, wherein, in response to the user being identified or authenticated, the at least one processor is further configured to control one or more devices to operate according to a desired setting of the user.

37. The control panel of clause 27, wherein the security event is associated with activation of a door switch of a door located next to the control panel.

38. The control panel of clause 27, further comprising a Bluetooth radio, wherein the security event is associated with one or more Bluetooth devices being in range, wherein, in order to determine that the security event has happened, the at least one processor is further configured to use the Bluetooth radio to detect the one or more Bluetooth devices.

39. The control panel of clause 38, wherein the at least one processor is further configured to:
 use the one or more still images or videos to perform facial recognition; and
 determine, based on the facial recognition, whether one or more individuals associated with the one or more Bluetooth devices are present at a premises.

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40. The control panel of clause 38, wherein the control panel further comprises one or more microphones, wherein the at least one processor is further configured to:
 record an ambient noise by the one or more microphones;
 use the ambient noise to perform voice recognition;
 use the one or more still images or videos to perform facial recognition; and
 determine, based on the voice recognition and the facial recognition, whether one or more individuals associated with the one or more Bluetooth devices are present at a premises.

41. A control panel comprising:
 at least one camera or microphone;
 a display configured to provide a graphical user interface;
 at least one memory; and
 at least one processor coupled with the at least one memory, wherein the at least one processor is configured to:
 perform at least one of a voice recognition or a facial recognition, using the at least one camera or microphone; and
 present, on the display, the graphical user interface including features associated with an individual recognized according to at least one of the voice recognition or the facial recognition.

42. The control panel of clause 41, wherein, in order to perform at least one of the voice recognition or the facial recognition, the at least one processor is further configured to execute an artificial intelligence module locally at the control panel.

43. The control panel of clause 41, wherein, in order to perform at least one of the voice recognition or the facial recognition, the at least one processor is further configured to:
 send data captured by the at least one camera or microphone to a cloud system; and
 receive an outcome of the voice recognition or the facial recognition from the cloud system, responsive to sending the data.

44. A door/window displacement detector comprising a capacitive sensing circuit configured to indicate whether a door/window is open or close.

45. A control panel comprising:
 at least one primary speaker configured to broadcast a first sound toward a front or a side of the control panel; and
 a removable back speaker removably attachable to a back side of the control panel and configured to broadcast a second sound toward the back side of the control panel.

46. A control panel comprising:
 at least one memory including a diagnostic tool; and
 at least one processor coupled with the at least one memory, wherein the at least one processor is configured to:
 execute the diagnostic tool locally at the control panel to at least one of test, measure, or graphically map a signal strength of a wired or wireless connection between the control panel and at least one sensor configured to communicate with the control panel over the wired or wireless connection.

47. A control panel comprising:
 a short range communication radio;
 at least one memory; and
 at least one processor coupled with the at least one memory, wherein the at least one processor is configured to:

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receive a short range communication signal transmitted by a device, using the short range communication radio; and

automatically disarm the control panel responsive to recognizing the device as a known device.

48. The control panel of clause 47, wherein the short range communication radio comprises a Bluetooth radio, a Bluetooth low energy “BLE” radio, or a near-field communication “NFC” radio.

Referring to FIG. 6 and FIG. 7, in operation for short range communication disarming functionality, computing device 600 may implement at least a portion of one or more components in FIGS. 1-5 above, such as all or at least a portion of the control panel 106 in FIG. 1, and may perform method 700 such as via execution of disarming functionality component 1026 by processor 605 and/or memory 610. Specifically, computing device 600 may be configured to perform method 700 for performing an aspect of short range communication disarming functionality, as described herein. It should be noted that computing device 600, processor 605, and memory 610 may be the same or similar to computing device 1000, processor 1002, and memory 1004 as described above with respect to FIG. 5.

At block 702, the method 700 includes receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device. For example, in an aspect, computing device 600, processor 605, memory 610, disarming functionality component 1026, and/or receiving component 620 may be configured to or may comprise means for receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device.

For example, the receiving at block 702 may include the control panel 106 executing the receiving component 620 to receive, by a short range communication radio in the control panel 106 (e.g., second other radios 134 in the control panel 106), a short range communication signal transmitted by a user device 650 (e.g., a smartphone of a user that is approaching the premises 102). In one non-limiting aspect, for example, a short range communication radio of the control panel 106 (e.g., second other radios 134 in the control panel 106) may be used to determine whether a user device 650 (e.g., a mobile user device such as a smartphone) is in range (e.g., using a receiver chain, an antenna, etc. in a short range communication radio of the control panel 106).

At block 704, the method 700 includes determining, by the control panel, whether the device is a known device. For example, in an aspect, computing device 600, processor 605, memory 610, disarming functionality component 1026, and/or determining component 625 may be configured to or may comprise means for determining, by the control panel, whether the device is a known device.

For example, the determining at block 704 may include the control panel 106 executing the determining component 625 to compare identifying data obtained by a short range communication radio of the control panel 106 from the detected user device 650 with stored known device data (e.g., stored in the memory 610) to determine whether the detected user device 650 is a known device. In one non-limiting aspect, for example, when a short range communication radio of the control panel 106 (e.g., second other radios 134 in the control panel 106) detects the user device 650 in range, the control panel 106 may retrieve stored data of known/registered devices from the memory 610 and then compare the stored data with corresponding data received by the short range communication radio from the user device 650 to determine whether the user device 650 is a known

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device (e.g., a registered smartphone). In one non-limiting aspect, for example, the user device 650 may include cooperating software (e.g., software installed to assist with identification). In these aspects, the control panel 106 may challenge the user device 650, for example, with a byte sequence, which the user device 650 hashes with a shared secret and the user device’s hardware network address, e.g., a MAC address. If the MAC address is recognized, the control panel 106 performs the same hash and confirms that the outcome of the hash matches the one submitted by the user device 650. In another non-limiting example, the control panel 106 may periodically beacon a byte sequence over BLE, where the sequence changes each time. The user device 650 responds by hashing its MAC address with a shared secret and the beacons byte sequence, and beaconing the hash. The control panel 106 detects the beacon and, if the MAC address is recognized, repeats the hash with the most recent one or two byte sequences, to make sure a stale value was not being used. The matching would identify the user device 650. In one non-limiting aspect, for example, a new user device 650 may contact the control panel 106, and the control panel 106 displays to a user, e.g., via the first UI 122 or via a device with an already authenticated app or via a web browser connected to the virtual control panel 110 in the cloud system 120, the identity of the new user device 650 so that the user may approve pairing the new user device 650. The control panel 106 then provides the new user device 650 with an encrypted token (to be presented by the user device 650 to the control panel 106 to identify the user device 650) or a shared secret (to be used for identification of the user device 650 in the above example aspects).

At block 706, the method 700 includes automatically disarming the control panel responsive to recognizing the device as the known device. For example, in an aspect, computing device 600, processor 605, memory 610, disarming functionality component 1026, and/or disarming component 630 may be configured to or may comprise means for automatically disarming the control panel responsive to recognizing the device as the known device.

For example, the disarming at block 706 may include the control panel 106 executing the disarming component 630 to automatically disarm itself (e.g., disable one or more security features of the control panel 106 at the premises 106, such as disabling an intrusion detection functionality of the security/automation system 100) responsive to recognizing the user device 650 as a known device (e.g., a registered smartphone). In an aspect, for example, the control panel 106 may be automatically disarmed in response to a known user device 650 being in range as detected by second other radios 134 in the control panel 106. In an aspect, the disarming may include disabling all of the intrusion detection functionality of the control panel 106, or only disabling a part of the intrusion detection functionality, e.g., disabling one or more zones less than all of the enabled zones.

In one optional implementation, the short range communication radio comprises a Bluetooth radio. In one optional implementation, the short range communication radio comprises a Bluetooth low energy “BLE” radio. In one optional implementation, the short range communication radio comprises a near-field communication “NFC” radio. For example, in some aspects, the short range communication radio in the control panel 106 (e.g., second other radios 134 in the control panel 106) may be a Bluetooth radio, a BLE radio, a NFC radio, etc. In one non-limiting aspect, for example, when a BLE radio in the control panel 106 (e.g., second other radios 134 in the control panel 106) recognizes

a BLE signal from a known user device **650**, the control panel **106** is automatically disarmed.

In one optional implementation, the determining whether the device is the known device at block **704** of the method **700** may further include determining whether the device is registered with the control panel as the known device. For example, in an aspect, computing device **600**, processor **605**, memory **610**, disarming functionality component **1026**, and/or determining component **625** may be configured to or may comprise means for determining whether the device is registered with the control panel as the known device.

For example, the determining at block **704** may include the control panel **106** executing the determining component **625** to retrieve stored data of registered devices from the memory **610** and then compare the stored data with corresponding data received by the short range communication radio from the user device **650** to determine whether the user device **650** is registered with the control panel **106** as a known device. For example, in one non-limiting aspect, instead of arming/disarming the control panel **106** by determining the exact location of a user, the control panel **106** arms/disarms based on a short range communication radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) detecting that a BLE user device **650** of a user is within BLE range and that the BLE user device **650** of the user has been registered with the control panel **106**.

In one optional implementation, at block **701** the method **700** may further include registering a device with the control panel by pairing the control panel with the device using a Bluetooth radio in the control panel. For example, in an aspect, computing device **600**, processor **605**, memory **610**, disarming functionality component **1026**, and/or registering component **635** may be configured to or may comprise means for registering a device with the control panel by pairing the control panel with the device using a Bluetooth radio in the control panel.

For example, in aspects where the short range communication radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) comprises a Bluetooth radio, the registering at block **701** may include the control panel **106** executing the registering component **635** to register the user device **650** by pairing the control panel **106** with the user device **650** using the Bluetooth radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**). In one non-limiting aspect, for example, the user device **650** may be a user smartphone which is paired with the control panel **106** in the premises **102**, e.g., using a built-in Bluetooth radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**). In one non-limiting alternative or additional aspect, the built-in Bluetooth radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) may also be used to pair new security or home automation sensors (e.g., security devices **104**) with the control panel **106**. For example, in an aspect, pairing a sensor/device **104** may be performed from a mobile phone app by scanning the sensor/device QR code and sending the sensor details to the control panel **106** using Bluetooth.

In one optional implementation, at block **710** the method **700** may further include unlocking, by the control panel, an entrance of a premises responsive to recognizing the device as the known device. For example, in an aspect, computing device **600**, processor **605**, memory **610**, disarming functionality component **1026**, and/or unlocking component **640** may be configured to or may comprise means for unlocking, by the control panel, an entrance of a premises responsive to recognizing the device as the known device.

For example, the unlocking at block **710** may include the control panel **106** executing the unlocking component **640** to send a signal/command to a security device **104** such as a door lock at an entrance of the premises **102** to unlock the door lock responsive to a short range communication radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) detecting the user device **650** in range and further responsive to the control panel **106** recognizing the user device **650** as a known device (e.g., a registered smartphone). Accordingly, in one non-limiting aspect, an entrance of the premises **102** is unlocked in response to the known user device **650** being in range of a short range communication radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**).

In one optional implementation, unlocking an entrance of a premises at block **710** of method **700** may further include sending, by a Z-wave radio in the control panel, a Z-Wave command to a door lock at the entrance of the premises. For example, in an aspect, computing device **600**, processor **605**, memory **610**, disarming functionality component **1026**, and/or unlocking component **640** may be configured to or may comprise means for sending, by a Z-wave radio in the control panel, a Z-Wave command to a door lock at the entrance of the premises.

For example, the unlocking at block **710** may include the control panel **106** executing the unlocking component **640** to send a Z-Wave command to a door lock at the entrance of the premises **102** by a Z-wave radio in the control panel (e.g., second other radios **134** in the control panel **106**) to unlock the entrance of the premises **102**. In one non-limiting aspect, for example, when a user approaches a front or back door of the premises **102**, the built-in Bluetooth radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) may detect that a smartphone of the user (e.g., the user device **650**) is within range. In response to detecting that the smartphone of the user (e.g., the user device **650**) is within range, the control panel **106** may disarm and also send a Z-Wave command by a Z-Wave radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) to unlock a door of the premises **102**. Alternatively or additionally, in one non-limiting aspect, for example, after the control panel **106** disarms and sends a Z-Wave command by a Z-Wave radio in the control panel **106** (e.g., second other radios **134** in the control panel **106**) to unlock a back door of the premises **102**, if the user opens the back door of the premises **102**, a sensor on the back door may send a signal to the control panel **106** to indicate that the back door has been opened. In response, the control panel **106** may, for example, chime and play an audio message such as "Back door opened!"

In one optional implementation, at block **712** the method **700** may further include controlling operation of one or more devices at the premises at a premises, by the control panel, responsive to recognizing the device as the known device. For example, in an aspect, computing device **600**, processor **605**, memory **610**, disarming functionality component **1026**, and/or controlling component **645** may be configured to or may comprise means for controlling operation of one or more devices at the premises, by the control panel, responsive to recognizing the device as the known device.

For example, the controlling at block **712** may include the control panel **106** executing the controlling component **645** to send a signal to one or more devices at the premises **102** to control their operation responsive to recognizing the user device **650** as a known device. For example, in some aspects, responsive to recognizing the user device **650** as a

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known device, the control panel 106 may operate based on a corresponding user configuration that may be saved (e.g., in memory 610) based on user inputs and/or based on an ML model that learns user behavior based on historical user actions. In some aspects, for example, the control panel 106 may control one or more lights at the premises 102 to turn them on responsive to recognizing the user device 650 as a known device, for example, based on a user preference/habit/historical pattern indicating to turn those lights on when the user arrives at the premises 102. In one non-limiting aspect, for example, when a user approaches a front or back door of the premises 102, a built-in Bluetooth radio in the control panel 106 (e.g., second other radios 134 in the control panel 106) may detect that a smartphone of the user (e.g., the user device 650) is within range. In response to detecting that the smartphone of the user (e.g., the user device 650) is within range, the control panel 106 may disarm and turn on the lights at the premises 102, for example, based on a user preference/habit/historical pattern indicating to turn those lights on when the user arrives at the premises 102. In alternative or additional aspects, responsive to recognizing the user device 650 as a known device, and based on an associated user preference/habit/historical pattern stored in the memory 610, the control panel 106 may turn on/off lights, update an operational setting (e.g., turn on/off, change temperature, etc.) of a heater or an air conditioner at the premises 102, turn on a sound system/music at the premises 102, turn on a video player/TV at the premises 102, turn on a faucet/shower at the premises 102, or perform any other home security or automation functionality in the security/automation system 100, e.g., according to user preferences/settings.

Some further example aspects are provided below.

1. A method comprising:

receiving, by a short range communication radio in a control panel, a short range communication signal transmitted by a device;

determining, by the control panel, whether the device is a known device; and

automatically disarming the control panel responsive to recognizing the device as the known device.

2. The method of clause 1, wherein the short range communication radio comprises a Bluetooth radio.

3. The method of any of the above clauses, wherein the short range communication radio comprises a Bluetooth low energy "BLE" radio.

4. The method of any of the above clauses, wherein the short range communication radio comprises a near-field communication "NFC" radio.

5. The method of any of the above clauses, wherein determining whether the device is the known device comprises determining whether the device is registered with the control panel as the known device.

6. The method of any of the above clauses, wherein the short range communication radio comprises a Bluetooth radio, the method further comprising registering the device with the control panel by pairing the control panel with the device using the Bluetooth radio in the control panel.

7. The method of any of the above clauses, further comprising unlocking, by the control panel, an entrance of a premises responsive to recognizing the device as the known device.

8. The method of any of the above clauses, wherein unlocking the entrance of the premises comprises sending, by a Z-wave radio in the control panel, a Z-Wave command to a door lock at the entrance of the premises.

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9. The method of any of the above clauses, further comprising controlling operation of one or more devices at the premises, by the control panel, responsive to recognizing the device as the known device.

5 An apparatus comprising:

a memory; and

a processor communicatively coupled with the memory and configured to perform the method of any of the above clauses.

10 A non-transitory computer-readable medium storing instructions executable by a processor that, when executed, cause the processor to perform the method of any of the above clauses.

15 An apparatus comprising means for performing the method of any of the above clauses.

The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more." The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term "some" refers to one or more. Combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words "module," "mechanism," "element," "device," and the like may not be a substitute for the word "means." As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase "means for."

What is claimed is:

1. A method comprising:

transmitting, by a short range communication radio in a control panel, a first beacon that comprises a byte sequence and is configured to cause a device that is within range of the control panel to generate a first hash of the byte sequence with a shared secret and a unique identifier of the device;

detecting, by the control panel, a second beacon transmitted by the device that is within range of the control panel, the second beacon comprising the first hash;

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determining, by the control panel, that the device that is within range of the control panel is a known device, responsive to recognizing the unique identifier of the device;

generating, by the control panel, a second hash of the byte sequence with the shared secret and the unique identifier of the device, responsive to recognizing the unique identifier of the device; and

automatically disarming the control panel responsive to determining that the first hash received from the device that is within range of the control panel matches the second hash generated by the control panel.

2. The method of claim 1, wherein the unique identifier comprises a Media Access Control (MAC) address of the device.

3. The method of claim 1, wherein the short range communication radio comprises a Bluetooth radio.

4. The method of claim 1, wherein the short range communication radio comprises a Bluetooth low energy (BLE) radio.

5. The method of claim 1, wherein the short range communication radio comprises a near-field communication (NFC) radio.

6. The method of claim 1, wherein determining whether the device is the known device comprises determining whether the device is registered with the control panel as the known device.

7. The method of claim 6, wherein the short range communication radio comprises a Bluetooth radio, the method further comprising registering the device with the control panel by pairing the control panel with the device using the Bluetooth radio in the control panel.

8. The method of claim 1, further comprising unlocking, by the control panel, an entrance of a premises responsive to recognizing the device as the known device.

9. The method of claim 8, wherein unlocking the entrance of the premises comprises sending, by a Z-wave radio in the control panel, a Z-Wave command to a door lock at the entrance of the premises.

10. The method of claim 1, wherein the device comprises a smartphone of a user associated with a premises monitored by the control panel.

11. The method of claim 1, further comprising using the short range communication radio to pair one or more security or automation sensors of a security or automation system with the control panel.

12. The method of claim 1, further comprising controlling, by the control panel, responsive to recognizing the device as the known device, an operation of one or more security or automation devices at a premises according to a stored user configuration, preference, habit, or historical pattern associated with the known device.

13. A control panel comprising:

one or more memories; and

one or more processors communicatively coupled with the one or more memories and configured to:

transmit, by a short range communication radio in the control panel, a first beacon that comprises a byte sequence and is configured to cause a device that is within range of the control panel to generate a first hash of the byte sequence with a shared secret and a unique identifier of the device;

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detect, by the control panel, a second beacon transmitted by the device that is within range of the control panel, the second beacon comprising the first hash;

determine that the device that is within range of the control panel is a known device, responsive to recognizing the unique identifier of the device;

generate, by the control panel, a second hash of the byte sequence with the shared secret and the unique identifier of the device, responsive to recognizing the unique identifier of the device; and

automatically disarm the control panel responsive to determining that the first hash received from the device that is within range of the control panel matches the second hash generated by the control panel.

14. The control panel of claim 13, wherein the unique identifier comprises a Media Access Control (MAC) address of the device.

15. The control panel of claim 13, wherein the short range communication radio comprises a Bluetooth radio.

16. The control panel of claim 13, wherein the short range communication radio comprises a Bluetooth low energy (BLE) radio.

17. The control panel of claim 13, wherein the short range communication radio comprises a near-field communication (NFC) radio.

18. The control panel of claim 13, wherein, in order to determine whether the device is the known device, the one or more processors are configured to determine whether the device is registered with the control panel as the known device.

19. The control panel of claim 13, wherein the short range communication radio comprises a Bluetooth radio, wherein the one or more processors are further configured to register the device with the control panel by pairing the control panel with the device using the Bluetooth radio in the control panel.

20. One or more non-transitory computer-readable media storing instructions executable by one or more processors, wherein the instructions, when executed, cause the one or more processors to:

transmit, by a short range communication radio in a control panel, a first beacon that comprises a byte sequence and is configured to cause a device that is within range of the control panel to generate a first hash of the byte sequence with a shared secret and a unique identifier of the device;

detect, by the control panel, a second beacon transmitted by the device that is within range of the control panel, the second beacon comprising the first hash;

determine that the device that is within range of the control panel is a known device, responsive to recognizing the unique identifier of the device;

generate, by the control panel, a second hash of the byte sequence with the shared secret and the unique identifier of the device, responsive to recognizing the unique identifier of the device; and

automatically disarm the control panel responsive to determining that the first hash received from the device that is within range of the control panel matches the second hash generated by the control panel.

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